

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel Group Study to Evaluate the Safety and Efficacy of KarXT + KarX-EC for the Treatment of Agitation Associated With Alzheimer's Disease **Short Title/Acronym:** ADAGIO-1 **Protocol Number:** CN012-0023 **NCT Number:** NCT07011732 **Trial Status:** Recruiting **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** KarXT + KarX-EC (Xanomeline-Trospium / Xanomeline Enteric Capsule, Xanomeline/Trospium Chloride Capsule) **Phase of Development:** Phase III **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead / Medical Director of Clinical Development

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of KarXT + KarX-EC in adult participants with agitation related to Alzheimer's Disease, specifically assessing the mean change from baseline on the Cohen-Mansfield Agitation Inventory-International Psychogeriatric Association (CMAI-IPA) total score at Week 14. **Secondary Objectives:** To evaluate changes in Clinical Global Impressions-Severity (CGI-S) scores, CMAI-IPA Domains, and CMAI Total Score, alongside additional broad safety, tolerability, and efficacy assessments.

2.2 Background & Rationale Agitation is a severe and highly prevalent neuropsychiatric complication affecting approximately half of all patients with Alzheimer's Disease. Despite its high clinical burden, treatment options remain extremely limited. Xanomeline-trospium pairs an M1/M4-preferring muscarinic receptor agonist with a peripherally restricted antagonist (trospium) to reduce cholinergic adverse effects. This muscarinic modulation mechanism provides a novel alternative to traditional dopamine receptor blockade, aiming to effectively manage behavioral and psychological symptoms of dementia (BPSD).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized, Parallel Group

3.2 Planned Enrollment & Duration Target \$N\$: 352 participants **Number of Sites:** Multicenter, Global **Estimated Study Start:** [DD-

MMM-YYYY] **Estimated Primary Completion:** 14-week treatment duration

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Ages 55 to 90 years. 2. Diagnosis of Alzheimer's disease (AD) in accordance with the 2024 Alzheimer's Association criteria, with biomarker confirmation of AD pathology (e.g., historical evidence of amyloid PET, A β 42/40 ratio in CSF, pTau181/A β 42 ratio in CSF, or pTau217/A β 42 ratio in plasma; or tested via plasma biomarker, amyloid PET, or CSF assays at screening). 3. Mini-Mental State Examination (MMSE) score of 5 to 22, inclusive, at Screening (Visit 1). 4. History of persistent agitation meeting the IPA consensus definition for agitation in cognitive disorders with onset at least 2 weeks prior to Screening (Visit 1). 5. NPI/NPI-NH Agitation/Aggression score ≥ 4 at Screening (Visit 1) and Baseline (Visit 2). 6. CGI-S score ≥ 4 , as related to agitation, at Screening (Visit 1) and Baseline (Visit 2). 7. At least 1 of the following 3 criteria must be established from the CMAI-IPA at Screening (Visit 1) and Baseline (Visit 2; CMAI-IPA Physical/Verbal Aggression Positivity): i) 1 or more aggressive behaviors occurring several times per week. ii) 2 or more aggressive behaviors occurring once or twice per week. iii) 3 or more aggressive behaviors occurring less than once per week. 8. Must have one identified caregiver who has sufficient contact (approximately 10 hours a week or more) and is willing to: attend all visits and report on participant's status, oversee participant compliance with medication and study procedures, participate in the study assessments, and provide informed consent.

4.2 Exclusion Criteria 1. Agitation symptoms that are primarily attributable to a medical condition other than the AD causing the dementia. 2. History of bipolar disorder, schizophrenia, or schizoaffective disorder. 3. History of (or at high risk for) urinary retention, gastric retention, or narrow-angle glaucoma as evaluated by the Investigator. 4. Risk of suicidal behavior during the study as determined by the Investigator's clinical assessment and/or C-SSR. 5. Recent history of receiving monoamine oxidase inhibitors, anticonvulsants (e.g., lamotrigine, divalproex), mood stabilizers (e.g., lithium), tricyclic antidepressants (e.g., imipramine, desipramine), or any other prohibited psychoactive medications except for as needed anxiolytics (e.g., lorazepam).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT + KarX-EC vs. matching Placebo. **Route of Administration:** Oral (Capsules)
Duration of Treatment: 14 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Selective serotonin reuptake inhibitors (SSRIs) and serotonin norepinephrine reuptake inhibitors (SNRIs) taken at a stable dose for at least 8 weeks prior to Screening (Visit 1). Mirtazapine or trazodone used as a hypnotic if started ≥ 8 weeks prior to Screening (Visit 1). As-needed anxiolytics (e.g., lorazepam). **Prohibited:** Monoamine oxidase inhibitors, anticonvulsants, mood stabilizers, tricyclic antidepressants, and other psychoactive medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -14 (Visit 1)	Informed Consent, Caregiver Verification, Biomarker Confirmation, MMSE, Clinical Agitation Assessments (NPI, CGI-S, CMAI-IPA)
Baseline	Day 0 (Visit 2)	Randomization, First Dose, Confirmation of Agitation Severity
Interim	Weeks 1 - 13	Safety Labs, Efficacy Assessments (CMAI-IPA, CGI-S), Caregiver Reporting
End of Study	Week 14	Final Efficacy Rating, Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean change from baseline on the Cohen-Mansfield Agitation Inventory-International Psychogeriatric Association (CMAI-IPA) total score at Week 14. **Measurement Method:** Clinical assessment tool evaluated by Investigator/Caregiver input.

7.2 Secondary & Exploratory Endpoints [Endpoint 1] Mean change from baseline on the Clinical Global Impressions-Severity (CGI-S) at Week 14. [Endpoint 2] Mean change from baseline on the CMAI-IPA Domains at Week 14. [Endpoint 3] Mean change from baseline on the CMAI Total Score at Week 14. [Endpoint 4] Additional safety and tolerability outcomes, and occurrence of cholinergic or other adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting via patient observation and caregiver interviews, specifically monitoring for anticholinergic/cholinergic effects and tolerability. **Serious Adverse Events (SAEs):** Reported by the investigator within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External committee established by the Sponsor (BMS).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy; Safety Set for all participants receiving ≥ 1 dose of the study drug. **Sample Size Justification:** Target enrollment of 352 participants designed to provide adequate power ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect statistically significant differences in CMAI-IPA and CGI-S mean change from baseline. **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) or Multiple Imputation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure integrity of clinician-rated scales and biomarker validation. **Audit Trail:** Validated eCRF tracking, secure data handling of caregiver-reported outcomes, and rigorous clinical scale rater training.

11. Study Identifiers & Governance

Primary ID: CN012-0023 **Posting ID:** 957664077916425312 **Registry Number:** NCT07011732 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** ADAGIO-1 **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel Group Study to Evaluate the Safety and Efficacy of KarXT + KarX-EC for the Treatment of Agitation Associated With Alzheimer's Disease

12. Clinical Framework (Agitation in Alzheimer's Specific)

Primary Condition: Alzheimer Disease (Agitation Associated With Alzheimer's Disease) **Diagnosis Threshold:** NPI/NPI-NH score ≥ 4 , CGI-S ≥ 4 , and CMAI-IPA positivity criteria met **Medical Methodology:** M1/M4-preferring muscarinic receptor agonism (Dopamine-sparing pathway) **Optimization Target:** Reduction in behavioral and psychological symptoms of dementia (BPSD)

13. Study Procedures & Methodology

Management Pathway: 14-week daily oral therapeutic intervention **Education Protocol (HCP):** Standardized training on rater scales (CMAI-IPA, CGI-S, NPI, MMSE) **Education Protocol (Patient):** Required primary caregiver involvement minimum 10 hours/week to support safety reporting, ensure compliance, and oversee efficacy

scaling **Quality Audit Mechanism:** Rater reliability checks and caregiver adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 14 Weeks (Treatment Period) **Onsite Visit Cadence:** Baseline, periodic assessments during the 14-week course, and End of Study. **Remote Monitoring Frequency:** Routine caregiver check-ins as defined by protocol. **Checklist Review Interval:** Continuous throughout the 14-week treatment.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					